

PLMS: A Packet Loss Memory-based Transmission Enhancement Scheme for GEO Satellite Networks

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Abstract—Geostationary earth orbit (GEO) satellites are vital components of 6G communication networks. However, GEO satellite networks face two main challenges. On the one hand, the long round-trip time makes the retransmission mechanism in the traditional transmission protocol slow to respond, which restricts the transmission performance. On the other hand, congestion packet loss caused by multi-stream competition and link packet loss due to high bit error rate coexist in GEO satellite networks, which makes it difficult for traditional congestion control (CC) algorithms to distinguish and make appropriate control decisions. To tackle these challenges, we propose a packet loss memory-based transmission enhancement scheme (PLMS). PLMS first modifies the TCP retransmission mechanism to accelerate the retransmission speed and enable the retransmission of packets that are lost again. Furthermore, PLMS introduces the packet loss memory mechanism to discern transmission link conditions, distinguish packet loss types, and dynamically adjust the sender's behavior. The mechanism can prevent misjudging bit error packet loss as congestion packet loss, thus improving CC accuracy. Our experimental results demonstrate that PLMS significantly boosts the throughput of existing CC algorithms. Specifically, under conditions of the 1% packet loss rate caused by bit errors, the throughput of CUBIC, BBR, and Hybla increase by 945%, 9.5%, and 154%, respectively.

Index Terms—GEO satellite network, congestion control algorithm, packet loss memory, transmission enhancement scheme

I. INTRODUCTION

Over several decades, geostationary earth orbit (GEO) satellites are widely utilized for various real-world applications [1]. Due to their wide-area coverage and reliable communication connection, GEO satellites play a critical role in providing ubiquitous communication services to ground users, especially in remote areas, and thus become an indispensable component of future 5G/6G networks [2], [3].

GEO satellites operate 35,786 km above the Earth's surface, and thus are severely affected by the weather [4]. This circumstance makes the satellite-ground communication link highly unstable and even get interrupted under extreme conditions [5], which poses significant challenges to end-to-end transmission. Specifically, transport protocols such as TCP, face transmission difficulties including high bit error rate (BER), long round-trip time (RTT), and asymmetric link bandwidth [4], [6]. These challenges severely compromise the performance of existing transport protocols, leading to GEO satellite networks failing to achieve their expected transmission efficiency [2].

Due to the unique challenges existing in GEO satellite networks, existing studies introduce ground proxy gateways [2], [7], which act as intermediaries, to isolate the GEO satellite network from the ground network and deploy transport protocols tailored to each network. Thus, with the development of satellite networks, new transmission enhancement scheme tailored for satellite networks continue to emerge [8]. Although new solutions enhance some communication capabilities of GEO satellite networks, they introduce many modifications that make it challenging to deploy. In terms of transmission protocols, LTP [9] is a cross-layer protocol that uses a proprietary congestion control (CC) algorithm, and its architecture is very different from the traditional transmission protocol, making it difficult to coordinate the transmission on the terminal. In terms of CC algorithms, AUTO [10] is a learning-based CC algorithm that requires additional computing resources from the gateway and is inefficient during the startup phase, exacerbating the impact of slow startup due to long RTT.

The use of typical TCP and its commonly used CC algorithms, such as CUBIC, BBR, and Hybla [11]–[13], makes the transmission between the gateway and terminals using different CC algorithms more coordinated and easier to deploy [9], but performs poorly in GEO satellite networks. The long RTT of the GEO satellite networks leads to a serious lag in the response of TCP retransmission mechanism to packet loss events, which results in severe queue congestion and queue backlogs [9]. The high BER of GEO satellite networks leads to CUBIC and Hybla to misjudge link packet loss as congestion packet loss, thereby reducing the transmission rate [8]. The high BER also leads to BBR's detection data to be severely damaged, affecting its judgment of network conditions. Therefore, designing a simple performance enhancement scheme tailored to the characteristics of GEO satellite networks, which is compatible with different CC algorithms to coordinate the transmission on different terminals used different CC algorithms, is highly effective and meaningful.

To address the main challenges in GEO satellite networks, e.g., high BER and long RTT, and further improve the performance of existing CC algorithms for end-to-end transmission in practice, we propose a lightweight scheme, packet loss memory-based transmission enhancement scheme (PLMS), applicable to common CC algorithms in GEO satellite networks. Specifically, the proposed PLMS addresses the aforementioned challenges from two perspectives: 1. PLMS

optimizes the retransmission mechanism of TCP based on the characteristics of GEO satellite networks, which can reduce the response time to packet loss information, and expedite retransmissions. 2. To address the high BER challenge, we introduce a packet loss memory (PLM) mechanism to distinguish packet loss types. Based on an accurate perception of the transmission link conditions, PLM employs different treatments for various packet loss types and dynamically adjusts the sender's behavior and *cwnd* size accordingly. Finally, we construct a GEO satellite network simulation platform based on ns-3 according to the characteristics of GEO satellite networks and evaluate the performance of PLMS. Results demonstrate that PLMS can significantly boost the transmission performance of existing CC algorithms in GEO satellite networks.

The contributions of this paper are summarized as follows:

- We propose a lightweight scheme, PLMS, to enhance the transmission performance of TCP while maintaining compatibility with existing CC algorithms, and thus provide a easy-to-deploy solution for proxy gateway-based GEO satellite networks.
- We propose two simple but effective mechanisms to boost the performance of existing CC algorithms. At first, retransmission timer mechanism can accelerate reaction speed and enable the retransmission of packets that are lost again. Second, PLM mechanism can distinguish various packet losses, and dynamically adjust the sender's behavior, which addresses the misjudgment problem triggered by high BER links.
- We construct a ns-3-based simulation platform and conduct extensive performance evaluations. Results demonstrate that PLMS significantly outperforms existing CC algorithms in terms of throughput while guaranteeing fairness in multi-flow conditions.

The remainder of this paper is organized as follows. In Section II, we introduce the GEO satellite network and the motivation behind the design of our scheme. In Section III, we propose the specific scheme design of PLMS. In Section IV, we conduct extensive performance evaluations compared with existing CC algorithms. Finally, conclusions are drawn in Section V.

II. MOTIVATION

The transmission environment of GEO satellite networks significantly differs from that of ground networks [1]. Therefore, GEO satellite service providers typically utilize ground proxy gateways to isolate satellite networks from ground networks [7]. As shown in Fig. 1, in this paper, we only consider the transmission segment between source gateway (GW) and destination GW. The entire transmission segment involving GEO satellite takes on a bent-pipe shape and is susceptible to the impact of variable weather conditions. According to the preliminary experiments, we conclude two primary motivations for this study, which are derived from our previous observations, as follows:

Inaccurate *cwnd* Variations: In the communication scenarios of GEO satellites, the high BER of GEO satellite

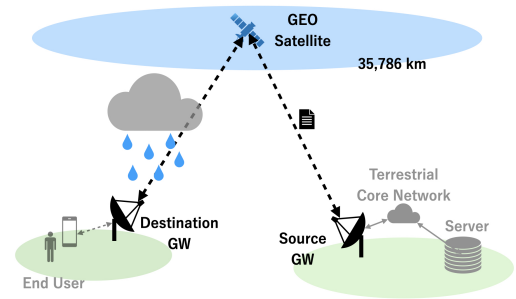


Fig. 1 Schematic diagram of the GEO satellite network.

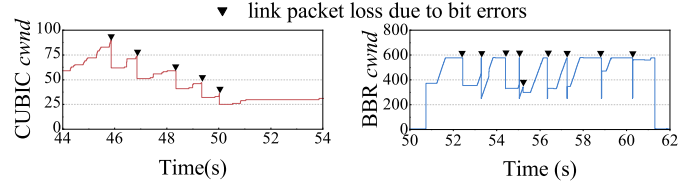


Fig. 2 *cwnd* of CUBIC and BBR in GEO satellite networks, bandwidth = 5Mbps, packet loss rate (due to bit errors) = 1%.

networks can lead to severe *cwnd* jitters at the sender. Fig. 2 illustrates the variations of *cwnd* using the CUBIC and BBR algorithms in GEO satellite networks. For the loss-based CUBIC, the sender incorrectly considers packet loss caused by bit errors as congestion, leading to a reduction of *cwnd*. The points where the CUBIC's *cwnd* decreases are all due to packet loss caused by bit errors. Even with detection-based BBR, frequent packet loss can lead to significant errors and jitters in BBR's detection information, resulting in significant *cwnd* fluctuations. Many algorithms that effectively address packet loss issues are primarily designed for ground networks or low earth orbit satellite networks [10], [14]. These algorithms, tailored for heterogeneous networks, are often complex, resource-intensive, and difficult to deploy. Moreover, in GEO satellite networks, packet loss caused by bit errors cannot be ignored, because it seriously interferes with the CC algorithms judgment of congestion packet loss and network conditions, limiting the performance of these algorithms. For GEO satellite networks characterized by simple network structures but complex channel environments, there is a lack of targeted, simple, and effective solutions.

Lightweight Extension Requirement: GEO satellite networks exhibit characteristics such as diverse service scenarios and many gateway connections [1]. Introducing a new single CC algorithm not only struggles to address the diversity of service scenarios but also entails extensive modifications to numerous gateways [15]. Moreover, the CC algorithm used by different terminals is also different and the gateway needs to coordinate the transmission with different terminals. Thus, in order to improve the quality of service for most GEO satellite networks, it is urgent to provide a lightweight performance enhancement scheme which is compatible with different CC algorithms and requires modifications solely at the gateway.

III. SCHEME DESIGN

In this section, we first optimize the TCP's retransmission mechanism for GEO satellite networks. After that, we introduce the PLM mechanism to perceive the packet loss condition of the link. Finally, the PLM mechanism classifies packet losses and precisely controls the behavior of the sender.

A. Retransmission Mechanism

The bandwidth-delay product (BDP) of GEO satellite networks is large, so a large number of packets are injected into the networks. If the receiver's queue is insufficient and a data packet is lost, it can lead to queue overflow, resulting in the loss of more subsequent packets and triggering a chain reaction collapse. Whether caused by congestion or bit errors, rapid response from the sender is necessary to reduce the backlog of network data packets. However, due to the long RTT of GEO satellite networks, the sender's response severely lags behind the real-time transmission status. Thus, rapid determination of packet loss and retransmission recovery can mitigate the influence of high delay caused by GEO satellite networks.

Due to the complexity of ground networks, packets may arrive at the receiver out of order. TCP avoids treating out-of-order packets as losses by setting the threshold for the retransmission recovery mechanism's duplicate ACK to 3. However, as shown in Fig. 1, the structure of GEO satellite networks is simple, typically comprising only three hops. Data packets are transmitted through the networks in a specific order and are not subject to disordering due to network heterogeneity. Thus, if a receiver receives an out-of-order packet, it is highly probable that the packets not received before this one are lost. In such cases, the sender does not need to wait for three duplicate ACKs. Instead, based on selective ACK (SACK), it promptly retransmits the lost packets.

Due to the high BER and BDP in GEO satellite networks, there is a probability that retransmitted packets may be lost again due to bit errors. TCP lacks effective mechanisms to handle the re-loss of retransmitted packets and typically resorts to waiting for timeout retransmissions, severely impacting transmission efficiency. The scheme proposes a retransmission mechanism that retransmits packets that have been lost multiple times. First, it introduces a retransmission timer (RTR). For each retransmitted packet, a RTR is set, which equals the current time plus a retransmission timeout (RT). The RT is set using exponential backoff based on RTT:

$$RT_i = 0.5 \times (RT_{i-1} + RTT_i). \quad (1)$$

When the current time exceeds the RTR, it is considered expired. Secondly, when the sender receives a SACK, it retransmits all packets that meet the following two conditions:

- Packets that have not been acknowledged before the SACK sequence number.
- Packets whose RTRs have expired.

The new mechanism ensures that the sender can retransmit data packets that are lost again without sending excessive redundant packets, which could degrade the link quality.

B. Packet Loss Memory

The high BER in GEO satellite networks significantly reduces the performance of many CC algorithms. To address this challenge, we introduce the PLM mechanism to perceive and record recent packet loss situations. PLM distinguishes between packet losses caused by bit errors and those caused by congestion, enabling different response behaviors to be adopted accordingly.

To perceive the condition of the GEO satellite link, we opt for the SACK mechanism, as it enables us to better acquire transmission information. When the sender receives SACK, we update a plm parameter. If the sender determines, based on the SACK, that data packets lose and necessitate retransmission, we increment the plm value:

$$plm_{i+1} = plm_i + re_{packets} \times plm_{scale}, \quad (2)$$

where $re_{packets}$ is the number of retransmitted packets, and plm_{scale} is the scaling factor. On the contrary, if the sender determines that data packet retransmission is not required, we decrease the PLM value:

$$plm_{i+1} = plm_i - 1. \quad (3)$$

C. Congestion Control

PLM mechanism integrates the plm parameter into existing CC algorithms to adjust their behavior and $cwnd$ size. Specifically, upon confirming packet loss, the sender implements different packet loss handling operations, such as how to reduce $cwnd$ and whether to adjust the transmission state based on different plm values.

- $plm \leq plm_{scale}$: In this scenario, when a packet loss is detected immediately after transmission, we attribute it to bit errors. For such occasional packet losses, PLM inhibits the original CC algorithm, refraining from reducing $cwnd$ and performing packet retransmission. Meanwhile, PLM treats the lost data packet as if it were received and executes corresponding actions, such as adjusting CC states and recording data.
- $plm_{scale} < plm \leq 2 \times plm_{scale}$: When two packet losses occur within a plm_{scale} window, we consider it to be a transitional phase from packet losses caused by bit errors to those caused by congestion. Thus, PLM slightly decreases the $cwnd$. Specifically, $cwnd$ decreases by the increment accumulated within one RTT of the current phase. This mechanism helps avoid repetitive probing of $cwnd$ and mitigates the impact of long RTT. A flow in the convergence phase enables the maintenance of high throughput over an extended duration.
- $plm > 2 \times plm_{scale}$: When a significant number of packet losses occur on the link within a short period, the link is likely congested. At this point, PLM activates the mechanisms of the embedded CC algorithm, returning to the original behavior logic of the CC algorithm. Meanwhile, let $plm = 0$ to prevent repeated entry into this state. Employing the embedded CC algorithm mechanism

helps prevent further exacerbation of congestion and ensures fairness among different connections.

Based on the above design, we know that the selection of plm_{scale} affects the effectiveness of packet loss differentiation. The value of plm_{scale} should not be too large. Otherwise, when the next packet loss occurs due to bit errors, plm has not decreased to 0 yet, leading to a misjudgment as congestion. The probability of misjudgment as congestion follows:

$$P_l = pbr^2 \times plm_{scale} \times (1 - pbr)^{(plm_{scale}-1)}, \quad (4)$$

where pbr is the packet lost rate due to bit errors. The value of plm_{scale} should not be too small either. Otherwise, when congestion starts to occur, plm will quickly drop to 0. The sender cannot detect congestion, leading to further deterioration of the link congestion. The probability of link deterioration follows:

$$P_c = (1 - pbr - pcr - pbr \times pcr)^{2plm_{scale}}, \quad (5)$$

where pcr is the probability of packet loss in congestion. Two misjudgment probabilities, P_l and P_c , are determined by two packet loss rates: pbr and pcr . Considering the case of GEO satellite networks such as 1% BER, plm_{scale} can be set in the range of 10-100 to ensure that the misjudgment probability is not higher than 1%.

D. Performance Analysis

Memory Overhead: A RTR for a lost packet consumes 64 bits of memory. In an extreme scenario where all packets are lost and a GEO satellite network is fully loaded with a BDP transmission, approximately 3000 packets per flow would require about 200 Kb of memory at most. plm needs only maintain one per flow, with a memory overhead of 32 bits.

Time Complexity: The additional processing time introduced by PLMS is relatively small. The time complexity for checking whether each retransmitted packet needs to be retransmitted again is $O(N)$. Additionally, PLM's classification processing has a time complexity of $O(1)$.

IV. PERFORMANCE EVALUATION

We first construct a realistic GEO satellite network environment on the ns-3-based simulation platform. Then, we conduct experimental evaluations of PLMS based on CUBIC, BBR, and Hybla. Due to the scarcity of CC algorithm disclosures by GEO satellite service providers, typical research is generally based on CUBIC, BBR, and Hybla [7], [9], [13].

A. Simulation Platform

To better assess PLMS's performance in the GEO satellite network environment, we construct a GEO satellite network simulation platform based on ns-3. Leveraging the characteristics of ns-3, we isolate each layer and primarily build modules below the network layer of the GEO satellite communication system. We develop motion models for GEO satellite nodes and ground gateways, a channel model for GEO satellite networks, as well as a transmission error model. Due to the relatively stationary nature of GEO satellites with respect to

the ground, ground gateways and GEO satellites adopt the stationary motion model.

The primary function of the channel model is to determine whether a connection can be established between two nodes based on their physical spatial positions. Additionally, the channel model abstracts the effects of external interference on transmission into variations in channel delay and bandwidth.

High BER is a significant characteristic of GEO satellite networks. Therefore, we implement a random packet loss model at the receiver to simulate dropping data packets that have experienced errors. The probability of random packet loss is denoted as:

$$pbr = 1 - (1 - P_b)^{size}, \quad (6)$$

where P_b is the probability of a bit being in error, and $size$ is the size of the packet.

B. Evaluation

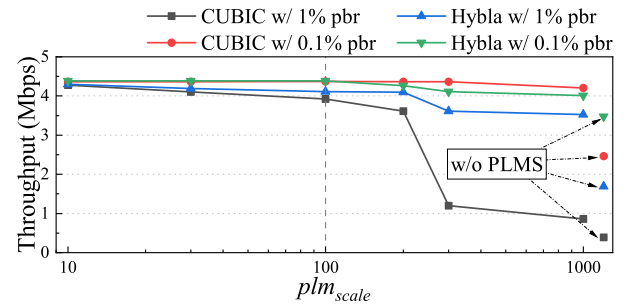


Fig. 3 Influence of plm_{scale} on PLMS.

In Fig. 3, we first evaluate the influence of different plm_{scale} parameters on CUBIC and Hybla. The control effectiveness of the plm_{scale} parameter is influenced by pcr and pbr . pcr is affected by the actual network environment and mainly determines the lower limit of plm_{scale} . To maintain consistency, we currently omit assumptions or examinations regarding pcr . Our primary focus is on how pbr affects the upper limit of plm_{scale} . At a 0.1% pbr , the tolerance upper limit of plm_{scale} is quite high. This is because even without integrating PLMS, the algorithms still meet acceptable performance standards. However, when the pbr is 1% and plm_{scale} is set to 300, the performance of CUBIC and Hybla obviously declines. This occurs because plm adds too much each time a packet is lost, accumulating to values that exceed the thresholds of the transition phase and congestion phase. Within the range of 10-100 for plm_{scale} , there is almost no difference in the PLMS's performance. As the plm_{scale} value increases, the PLMS's performance decreases until it becomes equivalent to the performance without PLMS. To adapt to more adverse transmission environments where pbr exceeds 1%, the subsequent experiments default to the plm_{scale} value of 30.

We conduct performance evaluations of the CUBIC, BBR, and Hybla algorithms, both with and without PLMS. Fig. 4 depicts the throughput variation over time. We simulate weather changes during the evaluation. Rainfall begins at 100

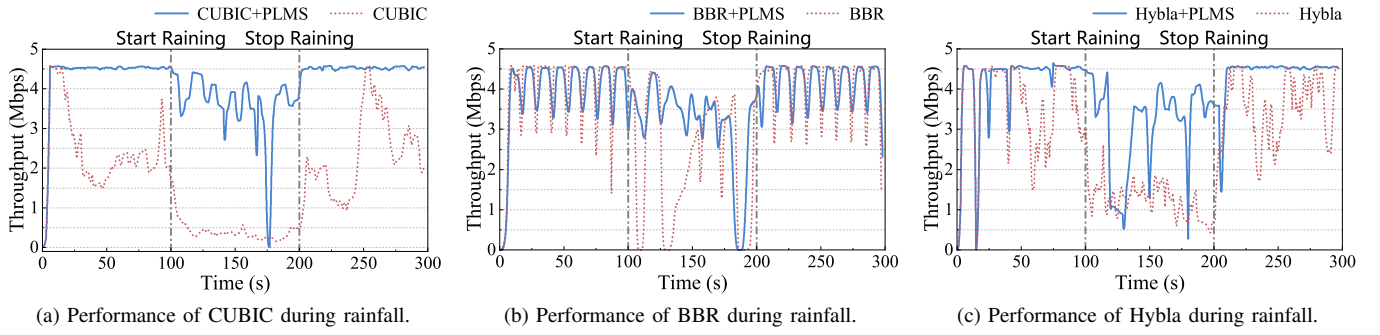


Fig. 4 Influence of weather changes on the CUBIC, BBR, and Hybla algorithms, both with and without PLMS.

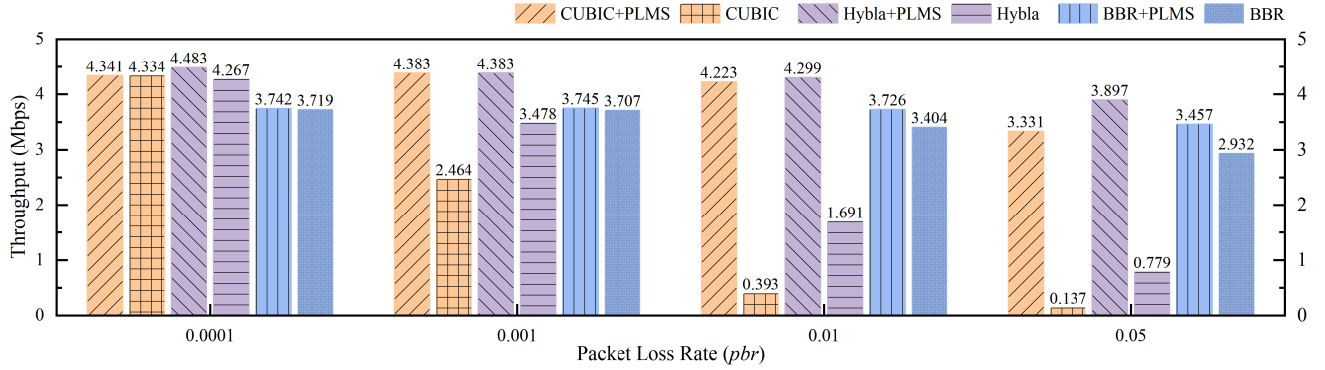


Fig. 5 Throughput of CUBIC, Hybla, and BBR algorithms under different packet loss rates, bandwidth = 5Mbps.

seconds and ceases at 200 seconds. During periods without rainfall, the link bandwidth is 5Mbps, and the link packet loss rate is 0.1%. During rainfall, the link bandwidth fluctuates between 3.2Mbps and 4.8Mbps, and the link packet loss rate fluctuates between 0.8% and 1.2%, while the other parameters remain constant. Fig. 4a illustrates the performance of CUBIC and CUBIC+PLMS. It can be observed that under a 0.1% packet loss rate, CUBIC achieves only half of the bandwidth throughput, and after rainfall, the throughput drops to less than 10% of the bandwidth. However, with the addition of PLMS to CUBIC, the throughput can maintain at least 90% of the bandwidth without significant fluctuations during non-rainfall periods. After rainfall, the throughput fluctuates with changes in channel conditions, but the average throughput still remains above 80% of the bandwidth. Fig. 4b illustrates the performance of BBR and BBR+PLMS. BBR demonstrates strong packet loss resilience, yet its throughput exhibits periodic declines. After rainfall, the link bandwidth changes, requiring BBR to re-detect the link's bottleneck bandwidth. However, due to the influence of long RTT, this process is very slow. The addition of PLMS shortens the time required for re-detection. Simultaneously, the sudden increase in packet loss rate, coupled with bandwidth contraction, results in the significant loss of packet information. It affects BBR's decision-making behavior, leading to instances of 0 throughput in the short term. Both the re-detection time and average throughput show significant improvements after integrating PLMS into BBR.

Fig. 4c illustrates the performance of Hybla and Hybla+PLMS. Due to Hybla's aggressive *cwnd* growth, its throughput exhibits considerable fluctuations. When transmitting steadily, the throughput stabilizes at around 90% of the bandwidth after integrating PLMS. During rainfall, PLMS can also maintain the average throughput above 70%.

In Fig. 5, we evaluate the performance of different algorithms under different packet loss rates. PLMS noticeably enhances CUBIC and Hybla's performance. The more severe the transmission environment, the more evident the improvements for both algorithms. Under a 1% packet loss rate, CUBIC's throughput increases by 945%, and Hybla's throughput increases by 154%. For BBR, due to limitations inherent in its algorithm mechanism, the improvement is relatively limited. With the addition of PLMS, BBR's performance increases by 10% under packet loss rates of 1% and 5%.

Table 1 Influence of PLMS on RTT.

Algorithm Name	CUBIC	Hybla	BBR
RTT w/ PLMS (ms)	509.7	523.1	542.6
RTT w/o PLMS (ms)	505.9	522.3	534.9
Ratio	1.008×	1.002×	1.015×

We evaluate the PLMS's influence on RTT. From Table 1, it can be observed that after incorporating PLMS, there is a slight increase in the RTT of each CC algorithm. PLMS ignores packet losses due to bit errors, which makes it easier

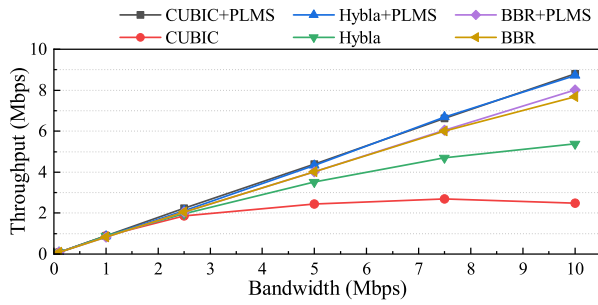


Fig. 6 Influence of bandwidth on PLMS, $pbr = 1\%$.

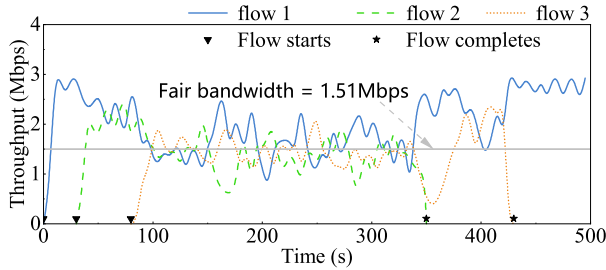


Fig. 7 Fairness of CUBIC utilizing PLMS.

to saturate the bandwidth and thus introduces queuing delays. However, due to the significant improvement in throughput achieved by PLMS, this slight increase in RTT is acceptable.

In Fig. 6, we evaluate the PLMS's performance under different bandwidths. When the bandwidth is low, the absolute number of packet losses is small, resulting in minimal influence on the performance of various algorithms, making it difficult to reflect the PLMS's performance. However, when the bandwidth exceeds 2.5Mbps, the PLMS's performance significantly improves, especially for CUBIC and Hybla. This is because, with higher bandwidth, there is more packet loss, which makes it more likely to encounter packet loss within one RTT. Thus, CC algorithms are more likely to enter the packet loss handling stage. PLMS can identify packet loss due to bit errors to prevent unnecessary *cwnd* reduction, thereby making full use of the link bandwidth.

In Fig. 7, we evaluate the PLMS's influence on fairness. We take the CUBIC algorithm as an example and set up three flows to evaluate the PLMS's fairness. Due to long RTT and large cache queues in GEO satellite networks, the three flows exhibit significant jitter. However, on average, PLMS can effectively ensure fairness among flows that use the same algorithm. Because congestion occurs, more packet loss occurs in a short period, which triggers the third condition of PLM. PLMS falls back to the original CC to ensure its fairness.

V. CONCLUSIONS

In this paper, we proposed a lightweight transmission performance enhancement scheme to improve the transmission quality of most GEO satellite networks. The basic idea was to optimize mechanisms for the long RTT and high BER challenges in GEO satellite networks while ensuring scheme

compatibility. For the long RTT, we optimized the TCP's retransmission mechanism to accelerate the sender's perception of the network. For the high BER, we proposed a new mechanism to differentiate packet loss types, enabling CC algorithms to more accurately control *cwnd*. Extensive experimental results demonstrated that our scheme significantly enhanced the transmission performance of common CC algorithms in GEO satellite networks.

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